Adopt a Pet

Analysis of what factors affect pet adoption

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Research Questions

1. Which factors have the greatest influence on pet's adoption?

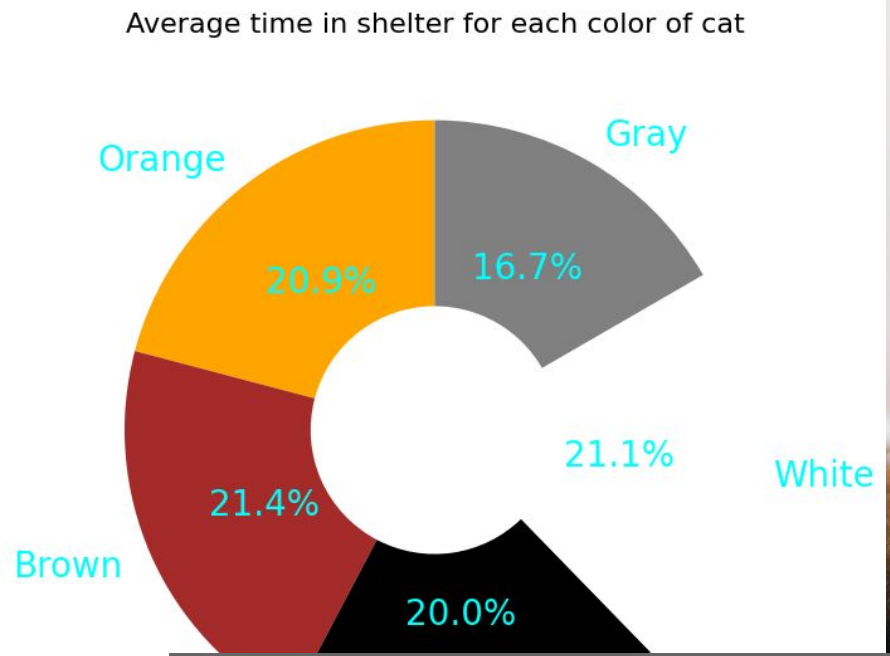
2. What is the average time each animal spends in the shelter?

3. How does color of the animal affect its chance of getting adopted based on species?

Color and Adoption Rates

Gray Cats spend the least amount of time in the shelter on average 36.6 day

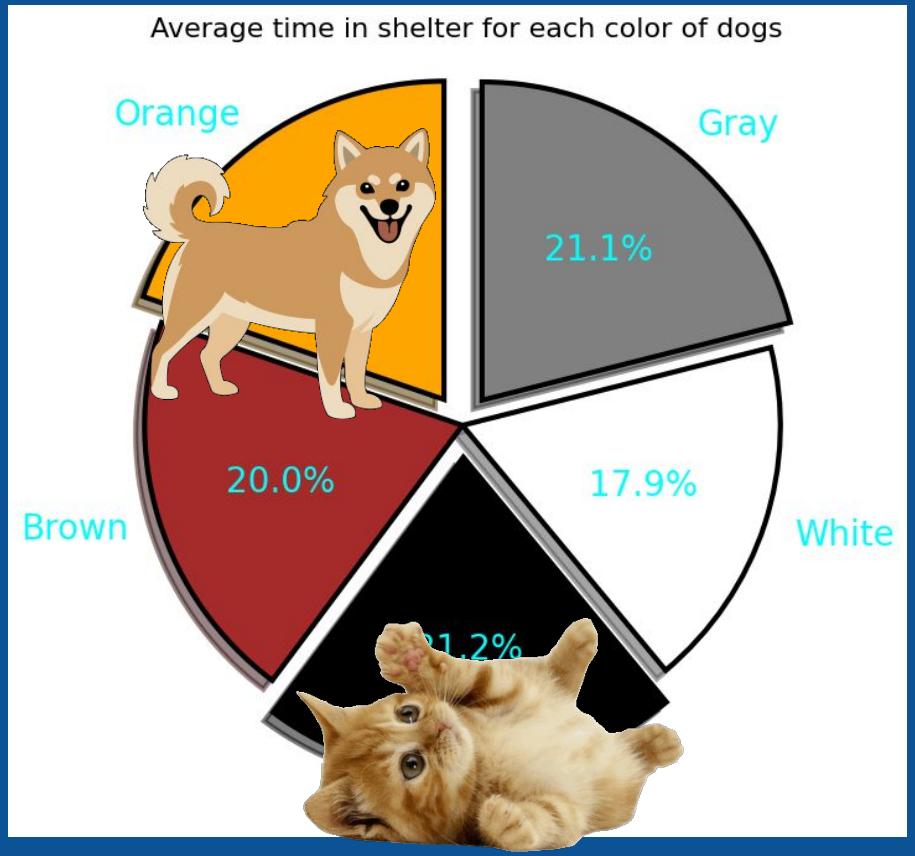
The other colors of cats spend 42 to 46 days in the shelter with black cats spending the least amount of time out of the remaining colors.

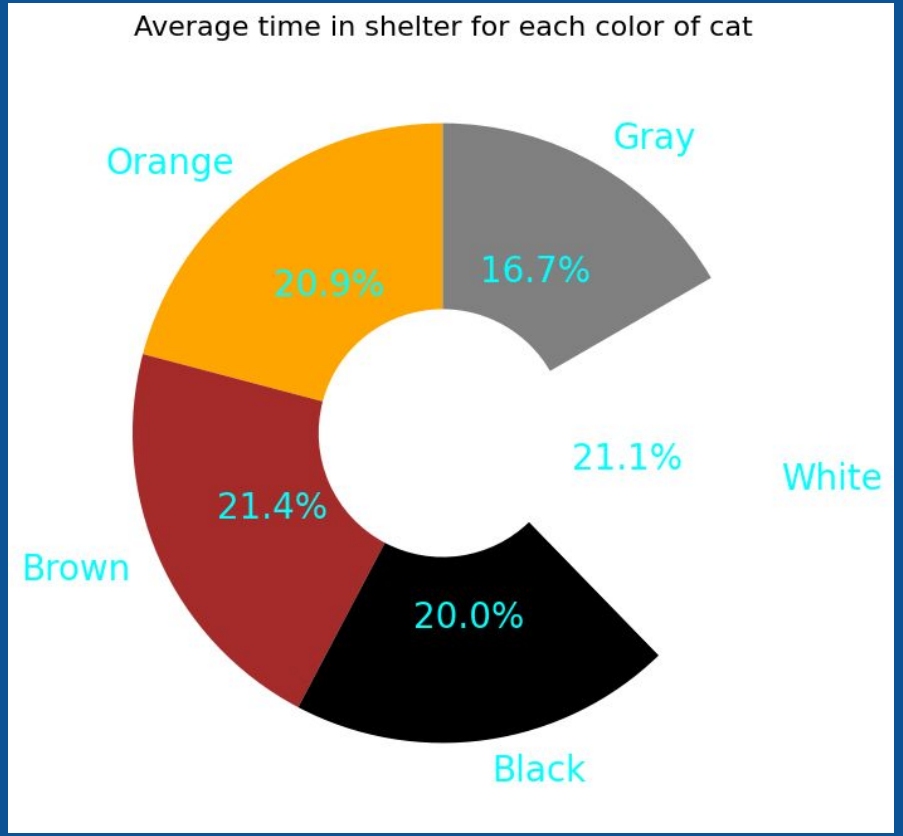


White Dogs spend the least amount of time in the shelter with an average of 39.6 day

The other colors of cats spend 44 to 47 days in the shelter with orange dogs spending the

Color and Adoption Rates

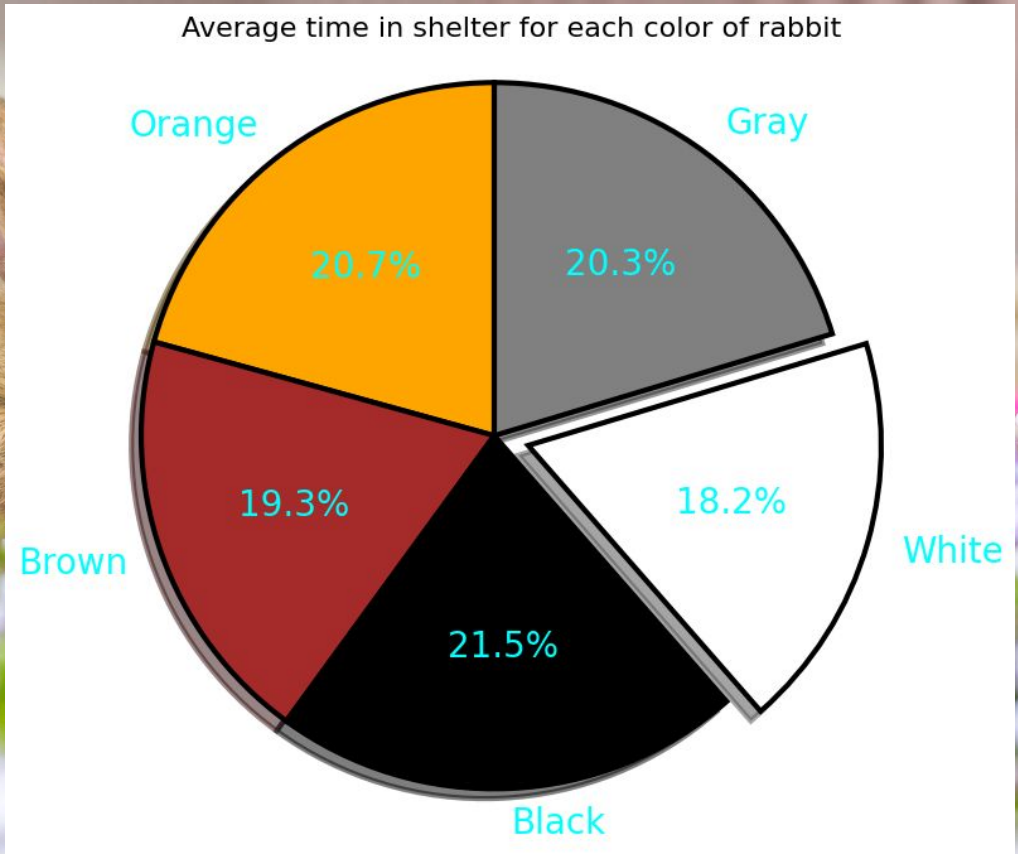


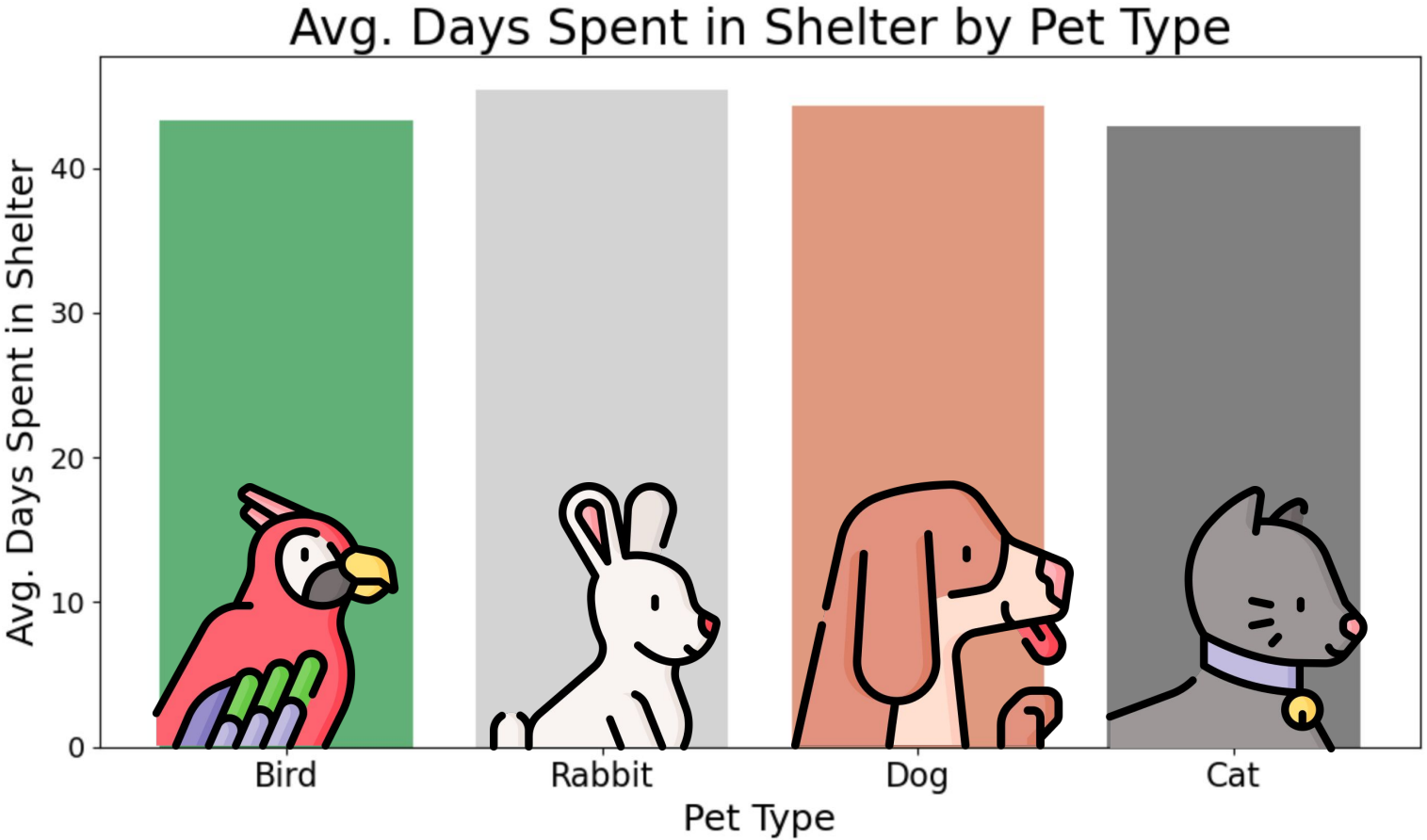


Color and Adoption Rates

White Rabbits spend the least amount of time in the shelter on average 41.2 day

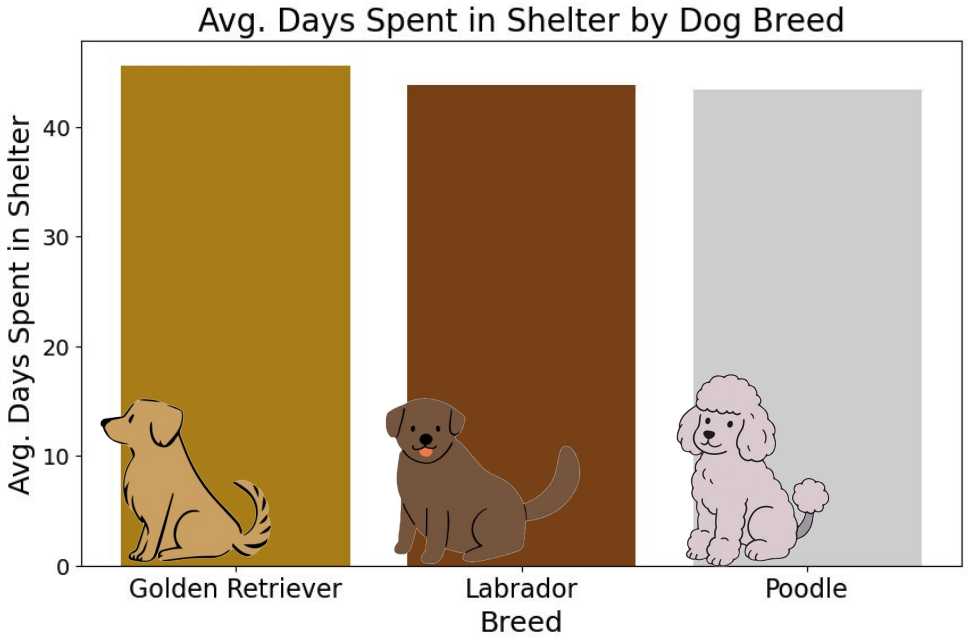
The other colors of cats spend 44 to 49 days in the shelter.

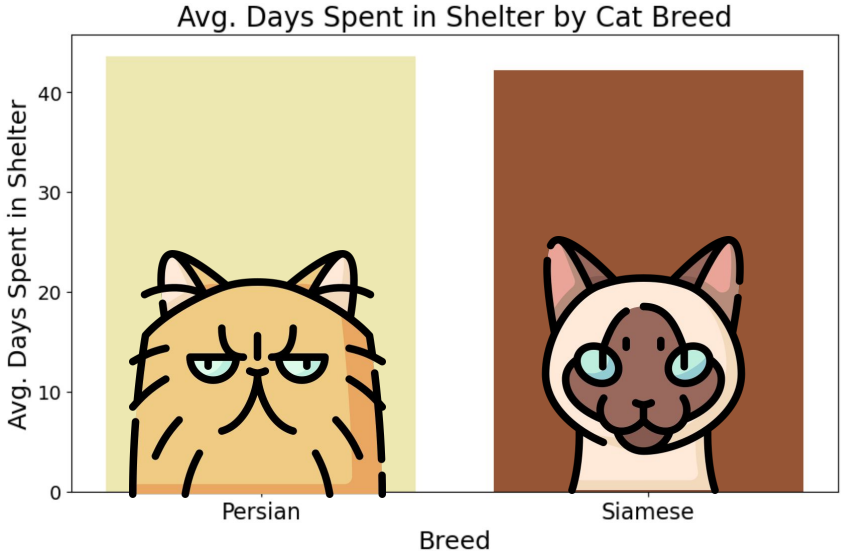




On average, rabbits spent the most days in the shelter (about 46 days), and cats spent the least (about 43 days).

Average Days Spent in Shelter by Breed

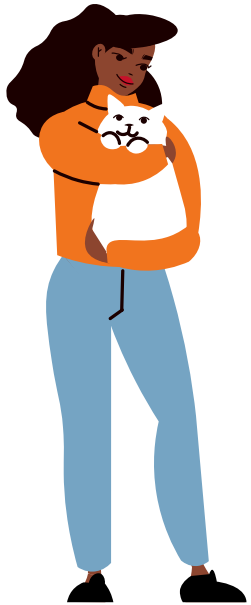




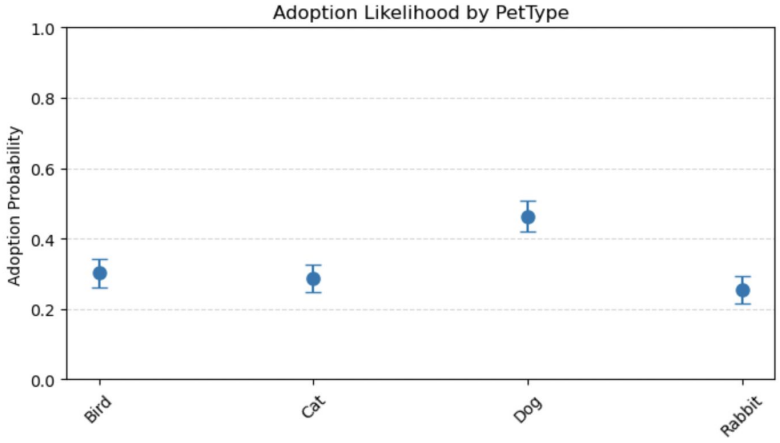
The highest avg. among dog breeds was 46 days for golden retrievers. Meanwhile for cats the highest average was seen in Persian cats at 44 days.

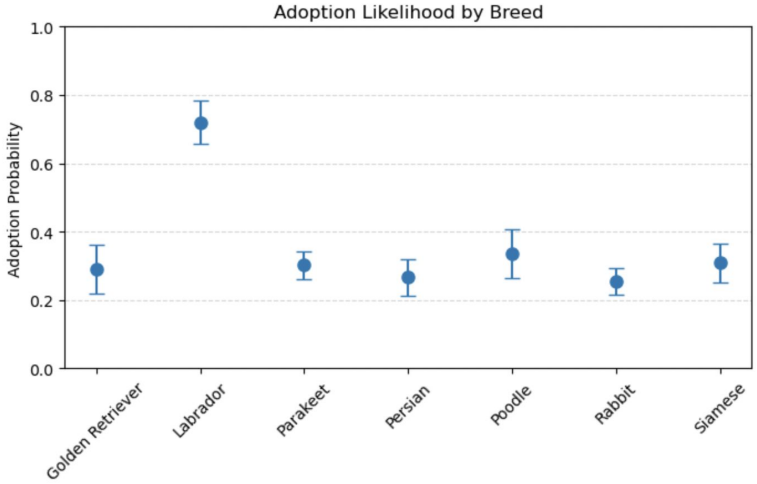
Adoption Likelihood by Factor

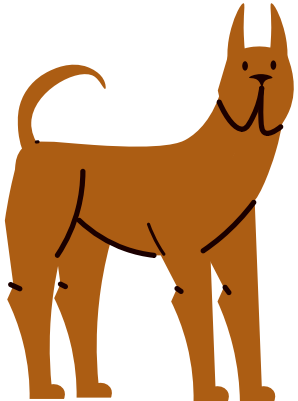
Factor	p value	Most Adopted
Age (Months)	p <0.005	Younger Pets
Weight (Kg)	0.60	
Adoption Fee	0.53	
Pet Type	p <0.005	Dogs
Breed	p <0.005	Labrador (Dogs)
Color	0.31	
Size	p <0.005	Medium

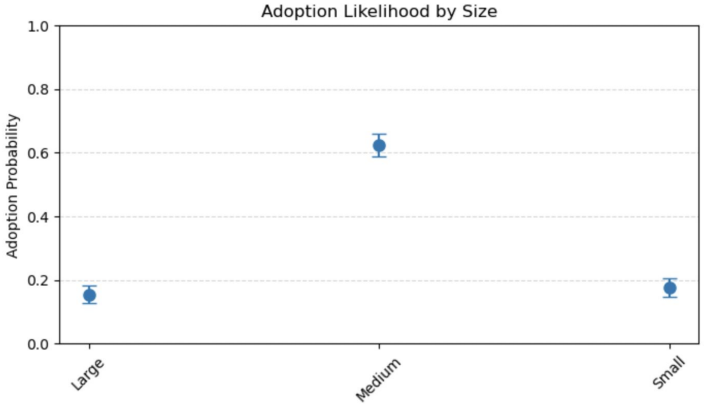


Adoption Likelihood by Significant Categorical Factors









Thank you!

Sources:

https://www.kaggle.com/datasets/rabieelkharoua/predict-pet-adoption-status-dataset

Jupyter Notebook, pandas, matplotlib, statsmodels.api, scipy.stats (chi2_contingency)